DESIGN AND ANALYSIS OF A STABILITY ENHANCED CMOS LOW DROP-OUT (LDO) REGULATOR

Uzair Nor Jasmin[1], Yusmeeraz Yusof[1]

[1] Malaysia - Universiti Teknologi Malaysia, Jalan Iman, 81310 Skudai, Johor Darul Ta'zim, Malaysia.

*Corresponding author:sp-uzair.nor.jasmin98@graduate.utm.my

ABSTRACT

This paper presents the design and analysis of a compensated CMOS Low-Dropout (LDO) regulator using an External Inserted Zero (EIZ) technique to improve loop stability without requiring a large external output capacitor. The proposed regulator targets automotive electronic systems including advanced driver-assistance systems (ADAS), in-vehicle infotainment, and electronic control units (ECUs) where stable and compact integration are critical requirements. The circuit consists of an error amplifier, PMOS pass transistor, feedback network, and compensation circuitry. Simulation was performed using LTspice to evaluate DC, AC, transient, and regulation performance and validated using MATLAB analytical modeling. The designed regulator achieved a phase margin of 90.77°, indicating stable operation under heavy and light load conditions. Quiescent current was maintained at 50 µA. Load regulation achieved 26.53 mV/mA, while line regulation was measured at 28.61 mV/V. The proposed design demonstrates effective stability enhancement suitable for automotive-grade capacitor-less LDO architectures and provides a basis for future optimisation toward AEC-Q100 compliance.

Keywords: Capacitor-less LDO; CMOS Low Dropout Regulator; Loop Stability; Line Regulation; Load Regulation